

IDP JSON ARTIFACT

Header

ID: IDP_0a4R30sjrEFes7mtu9Nbnv_003

Timestamp: 2024-03-15T14:32:07Z

"Evidence_data">

[IMG_PATH: /rgb/042.jpg] [THERM_PATH: /ir/042.tiff]

"Localization">

{global (GPS): [114.123, 22.384, 45.7m]}

{local: [x:0.09, y:2.79]m}

{relative: [0.46, 0.93]}

{semantic: "Top-Center"}

BIM_ID: IFC_WALL_0a4R30sjrEFes7mtu9Nbnv

"Analysis_results">

Stats

Count: 11

Defect:Crack

SEVERITY: CRITICAL

Material: Reinforced Concrete

Conf: 0.92

Digital Twin Platform Architecture

DIGITAL TWIN PLATFORM

IDP (Integrated Defect Profile)

Mapping 2D Masks to 3D Coordinates

- Geometric Reconstruction**
SfM / MVS
- Visual Perception**
Detection
- BIM/GIS Context**
Spatial Anchoring
- Topology Graph**
IFC Relations

Data Layer Modelling

- BIM/GIS Database**
IFC Models • Spatial Index
- Hierarchical Index**
Site → IAP → IDP
- Defect Records**
Imagery • Metadata
- Object Storage**
Images • BIM Files

COGNITIVE REASONING

Context Engine

- LLM Inference**
- RAG Retrieval**

Prompt Engineering

Knowledge Management

- Knowledge Graph Topology**
- Vector DB Embeddings**

Expert Knowledge (Standards)

Retrieves Context

Retrieval (Top-Down)

Interactions:

- Ingest:** Data flows from the top into the platform.
- Retrieve:** Data flows from the Context Engine into the platform.
- Instantiate:** Data flows from the platform into the IDP JSON artifact.